Niaz Bin Siraj

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Professional summary

I am seeking PhD opportunities in Computer Science, with a focus on AI for Software Engineering, reliable AI agents, and intelligent system design. My research motivation stems from 4+ years of hands-on experience architecting AI-powered production systems at Therap (BD) Ltd., where I have confronted open research challenges, including the reliability of LLM-generated database queries, security guarantees in natural-language-to-SQL translation, and the architectural implications of embedding AI agents into large-scale legacy codebases.

Professional experience

Software Engineer II *Jun 2022 - Present*
Therap (BD) Ltd. *Dhaka, Bangladesh*

- Investigated reliability of LLM-generated database queries in a production multi-tenant healthcare system, designing a zero-data-leakage NL-to-SQL layer serving thousands of US healthcare providers (Java 21, Spring Boot, DuckDB, Oracle).
- Developed an automated program comprehension pipeline using LLMs with incremental change detection for large-scale legacy codebases (Python, FastAPI, React, Oracle 26ai, Gemini).
- Designed and evaluated context engineering strategies for AI-assisted software development, reducing development time by 40% on a legacy monolith using MCP-connected knowledge retrieval.
- Engineered a history-based Aging Report Generation System improving accounts receivable accuracy across 30,000+ healthcare providers (Java, Spring, MyBatis, Oracle).
- Led the Nebraska State Integration Billing Flow from specification through deployment, ensuring regulatory compliance and 100% claim submission accuracy.
- Mentored 5+ junior engineers in system design, code review practices, and professional development.

Technical skills

Languages: Java 8/21, JavaScript, Python, SQL, HTML/CSS
Frameworks & Libraries: Spring, Spring Boot, MyBatis, Hibernate, FastAPI, React, JQuery
Databases & Messaging: Oracle, PostgreSQL, DuckDB, RabbitMQ, Redis, Coherence, HikariCP
Cloud & DevOps: Git, Bitbucket, Docker, Nginx, Jenkins, CI/CD, Gradle
AI & LLM Tools: LangGraph, RAG System, MCP, LLM Integration, Antigravity, Gemini CLI
Testing & Quality: JUnit, TestNG, Playwright, Postman

Selected projects

AI-Powered Natural Language Data Assistant | Java 21, Spring Boot, DuckDB, Oracle, LLM Integration

- Investigated reliability of LLM-generated SQL queries in a production multi-tenant healthcare system
- Designed a zero-data-leakage SQL security model enforcing strict tenant isolation under adversarial natural-language inputs
- Engineered sub-second analytical query execution over datasets spanning millions of healthcare claims

Context-Aware AI Development Workflow (MCP) | MCP, LangGraph, RAG, Python, Knowledge Graphs

- Designed and evaluated context engineering strategies for AI-assisted software development on legacy codebases
- Explored the architectural implications of MCP as a protocol for human-AI collaborative development workflows
- Demonstrated 40% reduction in development time through improved context retrieval and agent grounding

LLM-Powered Automated Code Documentation System | Python, FastAPI, React, Oracle 26ai, Gemini

- Developed an LLM-powered documentation generation pipeline with incremental change detection for large codebases
- Integrated vector embeddings (Oracle 26ai) for semantic search across auto-generated documentation
- Built a browser-based wiki interface for navigating generated documentation articles

Autonomous Maze Exploration and Shortest-Path Navigation Robot | Arduino, NodeMCU, Python, Flask, C#, Unity, Dynamic Programming

- Designed a backtracking-based maze exploration algorithm for autonomous robotic navigation
- Implemented BFS and dynamic programming optimization for shortest-path computation
- Built a real-time navigation simulation and visualization interface using C# and Unity

Education

University of Rajshahi *2017 - 2022*
Bachelor of Science in Computer Science and Engineering *Rajshahi, Bangladesh*

- CGPA: 3.30 / 4.00
- **Relevant Coursework:** Artificial Intelligence, Data Structures and Algorithms, Database Management Systems, Computer Networks, Operating Systems, Object-Oriented Programming, Software Engineering, System Design, Software Architecture, Web Technologies, Software Project Management
- Active member of the university Programming Club
- Represented university in ACM ICPC Dhaka Regional (2019, 2020, 2021)
- Volunteered at university-organized technology events

Research interests

AI for Software Engineering, Reliable and Trustworthy AI Agents, Natural Language Interfaces for Data Systems, Context Engineering for LLM-based Systems

Publications and presentations

[1] Niaz Bin Siraj, "Idea Showcasing with Poster Presentation at IC4ME2 2021," IC4ME2 2021, 2021.

[2] Niaz Bin Siraj, "Autonomous Maze Exploration and Shortest-Path Navigation for an Arduino-Based Intelligent Assistant Robot," Technical Report, 2022.

[3] Niaz Bin Siraj, "Research Paper Presentation - National Round, Children Science Congress," Children Science Congress, 2014.

Achievements and competitions

- Champion, National Round - Research Paper Presentation, Children Science Congress 2014
- Idea Showcasing - Poster Presentation at IC4ME2 2021, University of Rajshahi, Faculty of Engineering 2021
- ACM ICPC Dhaka Regional Onsite Finalist, ACM International Collegiate Programming Contest 2019
- Google Code Jam - Top 5901 out of 40,000+, Google 2020
- Certificate of Appreciation - Unity 3D Workshop, The American Center, U.S. Embassy Dhaka 2020
- ACM ICPC Dhaka Regional Online Preliminary - Rank 118, ACM International Collegiate Programming Contest 2020
- ACM ICPC Dhaka Regional Online Preliminary - Rank 137, ACM International Collegiate Programming Contest 2019
- ACM ICPC Dhaka Regional Online Preliminary - Rank 247, ACM International Collegiate Programming Contest 2021
- SRBD Coding Contest - Rank 89, SRBD Coding Contest 2018

Teaching and mentoring

Workshop Facilitator - Game Development with Unity 3D *2020*
The American Center, U.S. Embassy Dhaka

- Facilitated a hands-on workshop introducing participants to game development fundamentals using the Unity 3D game engine. Covered core concepts including physics simulation, scripting in C#, and iterative design principles.

Engineering Mentor *2023 - Present*
Therap (BD) Ltd.

- Mentoring 5+ junior software engineers in backend architecture patterns, code review best practices, system design thinking, and professional development. Providing structured guidance on complex problem decomposition and production-grade code quality.